PEARSON EDEXCEL INTERNATIONAL GCSE (9–1) · PHYSICS · 4PH1/2P

Rapid Revision Summary

Last-minute review · condensed from 17 papers and mark schemes, Jan 2019 – Jun 2026

THE PAPER

1 h 15 min · 70 marks · 7–9 questions · ≈ 1 min per mark. One long method (5–6 marks), multi-step calculations, short explanations. Whole spec incl. Paper-2-only content.

TOP 12 MARK-LOSERS

- No unit conversion (ms, cm, g, kPa, days) → **−1**.
- "m" for momentum; "C" for charge or current.
- "Momentum is conserved" alone.
- "Absorbs the impact" for safety features.
- Positive charges or "positive electrons" moving.
- Gas law in °C instead of **kelvin**.
- "Particles expand".
- Absolute magnitude without **10 parsecs**.
- "Mechanical energy" instead of transferred **mechanically**.
- "Scale" instead of **balance**.
- Anomaly in a mean; wrong precision.
- Unqualified "cheap/reliable/greenhouse gases".

EQUATIONS

Quantity	Equation
momentum	$p = mv$
force	$F = \Delta p \div t$ (given)
moment	$F \times d$
charge	$Q = It$
energy	$E = QV$
transformer	$Vp/Vs = Np/Ns$
power	$VpIp = VsIs$
heat	$\Delta Q = mc\Delta T$ (given)
gas (V const)	$p_1/T_1 = p_2/T_2$
gas (T const)	$p_1V_1 = p_2V_2$
frequency	$f = 1/T$
redshift	$\Delta\lambda/\lambda_0 = v/c$ (given)

MOMENTUM

- 72 × 13 = 936 kg m/s; F = 936 ÷ 0.29 = 3200 N.
- Explosion: 0 = 1.9 × 3.7 + 2.9v ⇒ v = −2.4 m/s.
- 39 ÷ 8.1 = 4.8 m/s; 39 ÷ 0.56 = 70 N.
- t = 57 000 ÷ 6100 = 9.3 s.

SAFETY (3)

Longer impact time → smaller rate of change of momentum → smaller force.

MOMENTS

- 28 × 0.15 = 4.2 N m. 230 × 0.37 = 0.98F ⇒ F = 87 N.
- Clockwise = anticlockwise about one point.
- Bigger effect: bigger force or longer handle.
- Load moves to B ⇒ force on B ↑, A ↓; sum = total weight.
- Centre of gravity: where weight appears to act.

ELECTROSTATICS AND TRANSFORMERS

CHARGE

Friction → electrons transferred; loses electrons = positive.

Earthing: electrons flow through wire → neutral → no spark/fire.

Like repel, unlike attract.

- $Q = It$: $6.8 \times 10^{-6} \times 3.7 \times 10^{-3} = 2.5 \times 10^{-8}$ C. $V = E/Q = 3.7/0.0043 = 860$ V.
- Electron $v = \sqrt{(2KE/m)} = 5.3 \times 10^{7}$ m/s.
- Electromagnet: coil + iron core + current.
- $40/N_s = 6.8/9.9 \Rightarrow N_s = 58$. $275\,000\ I_p = 230 \times 95 \Rightarrow I_p = 0.079$ A.

TRANSFORMER (4) AND GRID (5)

a.c. in primary → changing field → iron core → induces a.c. voltage in secondary.

Step-up: V ↑, I ↓ → less heating loss in cables. Step-down near homes for safety.

WAVES AND SOUND

OSCILLOSCOPE FREQUENCY (5)

Microphone → steady trace, ≥1 cycle → squares × time base = T → f = 1/T. 4 × 0.25 ms = 1.0 ms ⇒ 1000 Hz.

- Speed of sound: echo (≥50 m, double distance) or see-and-hear (≥100 m); tape measure, stopwatch, repeat.
- 180 ÷ 0.55 = 330 m/s; error: reaction time.
- Hearing 20 Hz–20 000 Hz; λ = 330 ÷ 25 000 = 0.013 m.
- Doppler: shorter λ ahead (higher f).
- Radar: distance = vt ÷ 2.

ENERGY RESOURCES

Resource	+	−
solar	renewable, no CO_2	night, space
hydro	no fuel, fast response	flooding
nuclear	no CO_2, reliable	waste, accidents
fossil	reliable	CO_2, SO_2, finite

- Hydro: GPE store → kinetic (mechanically) → turbine → generator → electrically.

- Nuclear: nuclear store → thermal store of coolant → kinetic of turbine.
- P = E/t: $9.7 \times 10^{14} \div 86\,400 = 1.1 \times 10^{10}$ W.

THERMAL PHYSICS AND GASES

SHC EXPERIMENT (6)

Balance (mass) → start T → heater, record V, I, t (E = VIt) → stir → max T → c = E/(mΔT) → repeat; insulate + lid. Heat losses make c too high.

$54\,000 \div (0.56 \times 22) = 4400 \cdot 3744 \div (0.29 \times 28) = 460$ J/kg °C.

- $0.016 \times 4200 \times 18 = 1210$ J.
- Solid: regular, vibrate. Liquid: close, irregular, slide. Gas: far, fast, random.
- Melting: bonds weaken; temperature constant (plateau).
- Hotter gas → faster particles → more frequent, harder collisions.
- $136 \times 230/400 = 78$ kPa (kelvin). Absolute zero −273 °C.
- Trapped air: poor conductor, no convection.

NUCLEAR

- Fission: neutron → U-235 splits → daughters + 2–3 neutrons → chain.
- Fusion: high T (high KE to overcome repulsion of positive nuclei) + high p (close, frequent collisions).
- Alpha repelled ⇒ nucleus positive.

ASTROPHYSICS

HR DIAGRAM

Absolute magnitude vs colour/temperature (hot left). Main sequence diagonal; red giants top right; white dwarfs bottom left.

Absolute magnitude: brightness at 10 parsecs.

BIG BANG

Evidence: CMBR + red-shift of galaxies.

Further galaxies → bigger red-shift → faster → universe expanding.

CMBR same everywhere; wavelength stretched as universe cooled.

$v = 3.0 \times 10^8 \times 15/630 = 7.1 \times 10^6$ m/s.

METHODS

- Ball bounce vs T: water bath, range of T, eye-level height, same drop, repeat.
- Black vs silver: same volume, distance, start T; fixed time; repeat.
- Density: zeroed balance, displacement can/cylinder, ρ = m/V with unit.